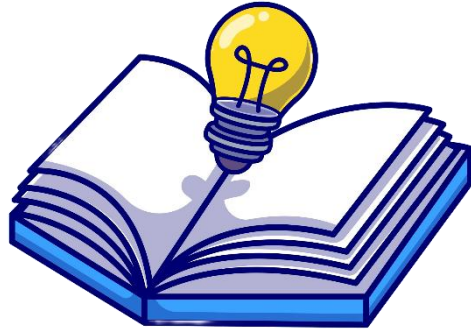


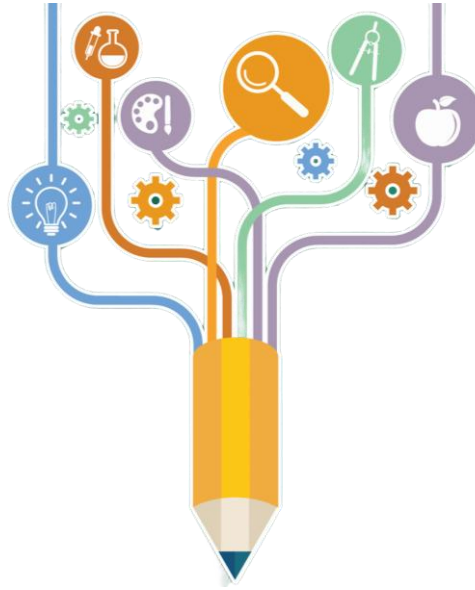
Project and Design Portfolio

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Welcome, and thank you for taking the time to view my portfolio. The goal of this portfolio is to give you a deeper insight into my experiences and skills I have gained over my recent history



It is my hope that this will allow you to better assess how my skills can be applied to your company.



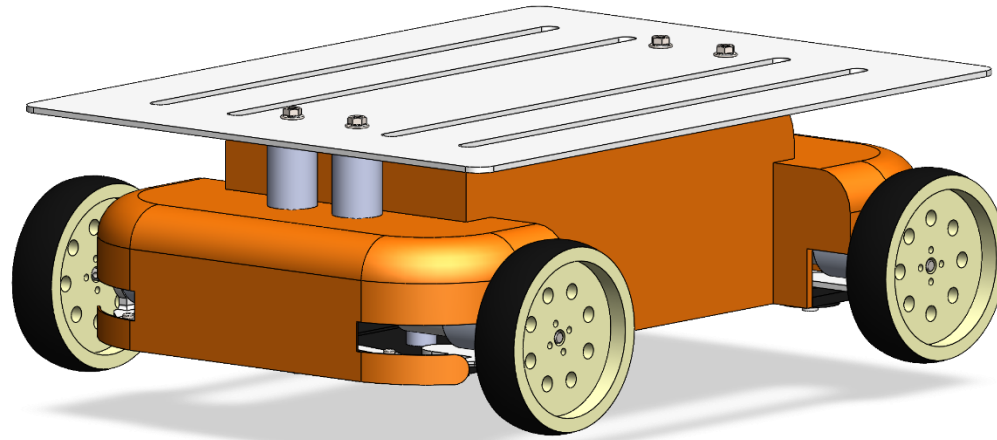
I would be happy to talk in more detail and can be reached using the contact information on the previous page.

Project Title: Industrial Kuli: Synchronized Crab-Steering AGV

A single-actuator omni-directional vehicle designed for precision logistics in tight industrial spaces.

Problem Statement:

Existing industrial Automated Guided Vehicles (AGVs) often require expensive, complex multi-motor systems to achieve crab steering and manoeuvring in narrow warehouse aisles.

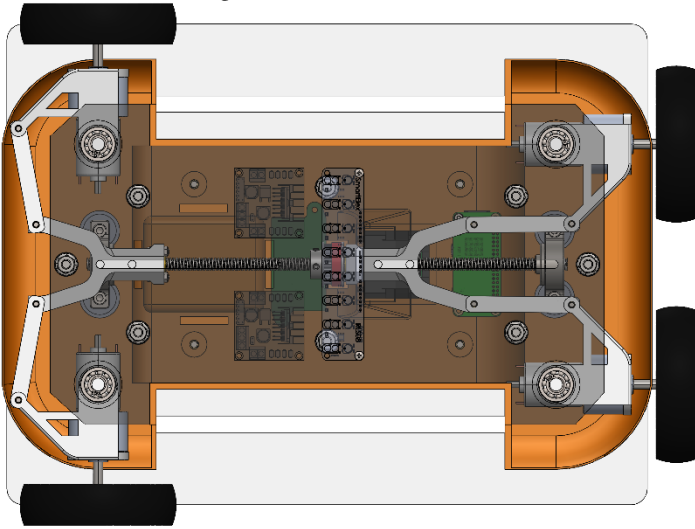


Key Achievement:

Successfully achieved mechanical synchronization of 4-wheel steering angles with <2% error variance using a custom dual-nut lead screw drive.

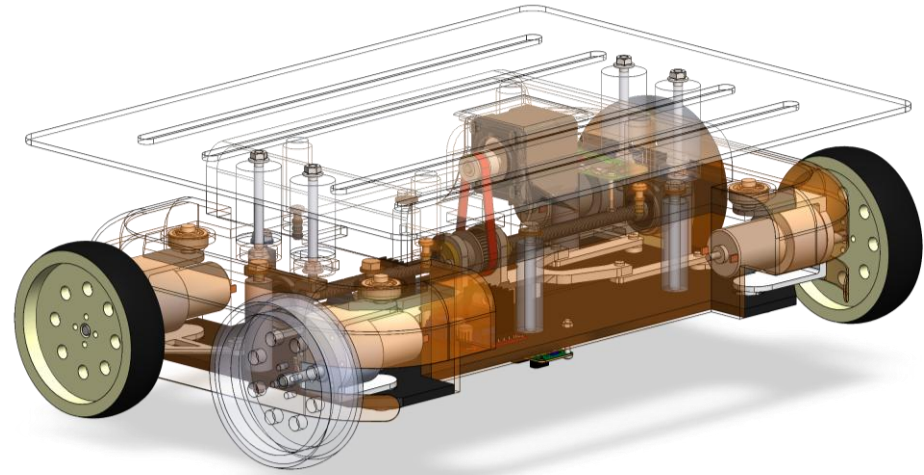
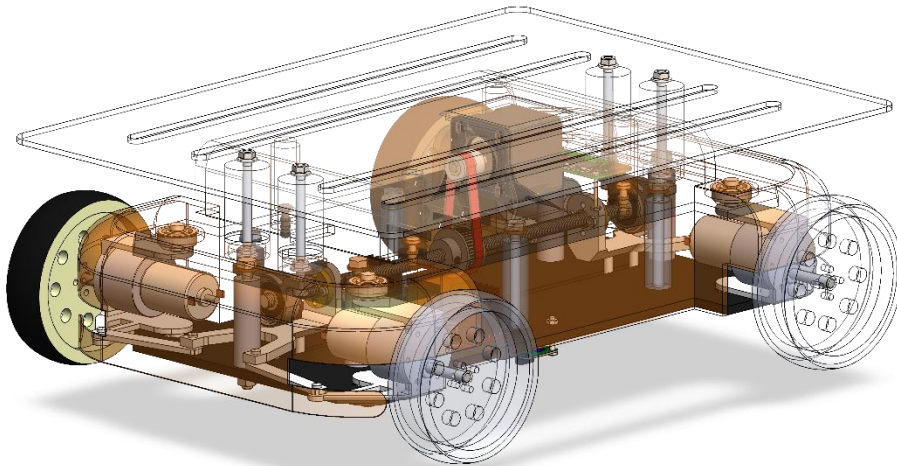
Solution/Impact: The "Kuli" reduces the actuator count by 75% compared to standard 4-wheel independent steering systems, lowering manufacturing costs while maintaining high-precision manoeuvrability for confined industrial environments.

Mechanical Synchronization & Hybrid Navigation



About Features:

- **Single-Actuator Steering:** A central lead screw drives two opposing copper nuts. These nuts push/pull Y-linkages that pivot the wheel hubs, forcing perfect mechanical symmetry in steering angles without complex software feedback loops.
- **Hybrid Guidance:** Combines low-level magnetic tape following for robust path adherence with high-level 2D map planning for route optimization.



Achievement: Validated stable material transport at speeds of 0.3–0.5 m/s while executing lateral (crab) translation.

Project Title: Hootie: Biomimetic IoT Companion Robot

An interactive desktop robot blending animatronic biomimicry with IoT connectivity.

Problem Statement:

Human-Robot Interaction (HRI) often feels sterile; this project aimed to create an emotive, "alive" interface for standard IoT tasks like notifications and timekeeping.



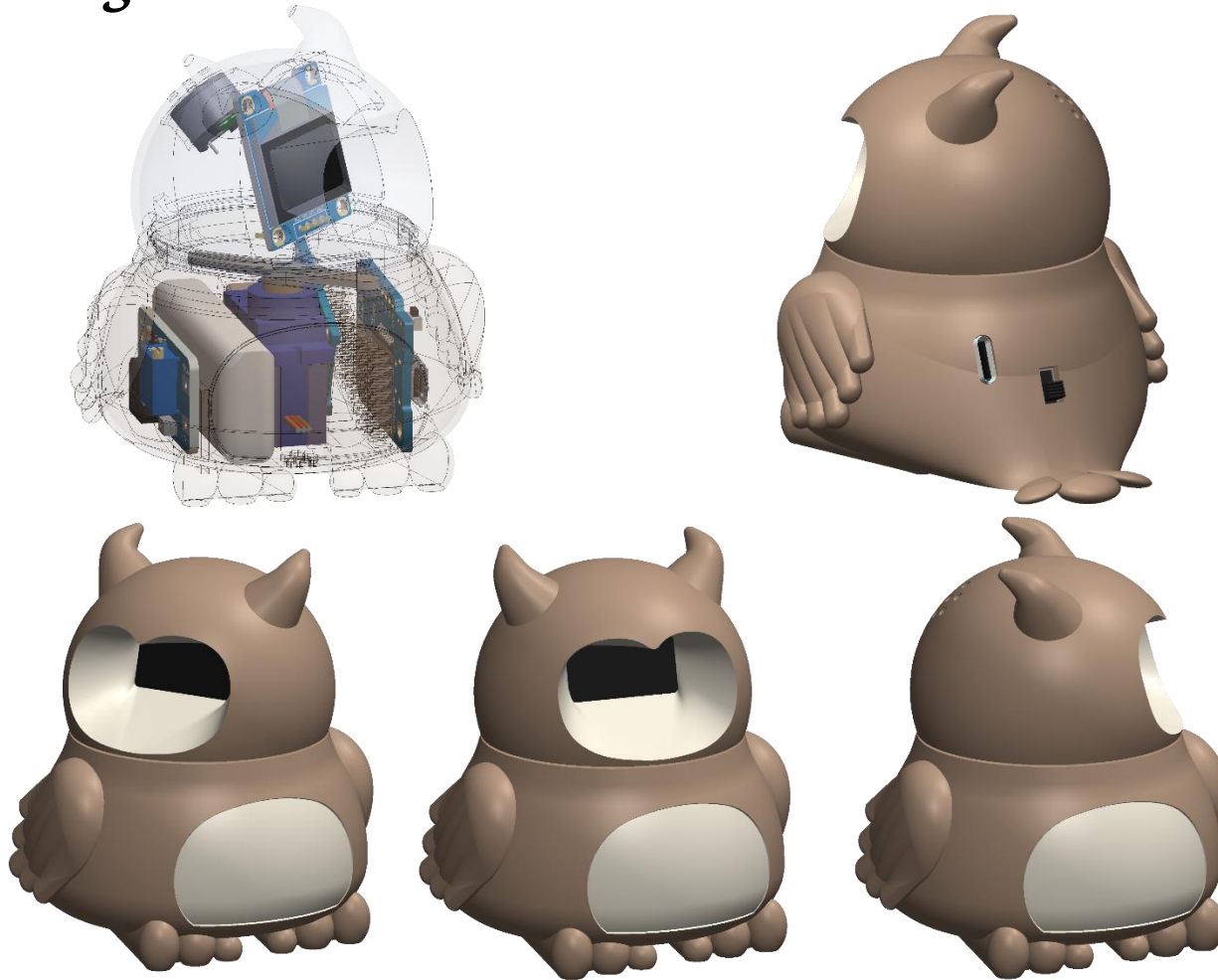
Hootie is a 3D-printed desktop companion that functions as an autonomous digital pet and an internet-connected clock. It features a custom "Chimney Effect" passive cooling system, a screwless snap-fit enclosure, and animate OLED eyes that react to user input and server commands.

Key Achievement:

Developed a multi-modal C++ state machine allowing seamless transitions between "Autonomous Life," "Puppeteer Mode" (Remote Control), and "Utility Mode" (Clock).

Solution/Impact: Successfully demonstrated how biomimetic movement (saccadic eye motion, randomized blinking) improves user emotional engagement with IoT devices.

Integrated Mechatronics & Web Server Control



About Features:

- **Screwless Assembly:**
Designed a fully snap-fit enclosure inspired by consumer electronics, utilizing locking head mechanisms to prevent servo strain.
- **IoT "Puppeteer" Mode:**
The ESP8266 hosts an asynchronous web server, allowing users to manually control the robot's servo motors and trigger emotion presets (Love, Anger, Confusion) via a smartphone browser.

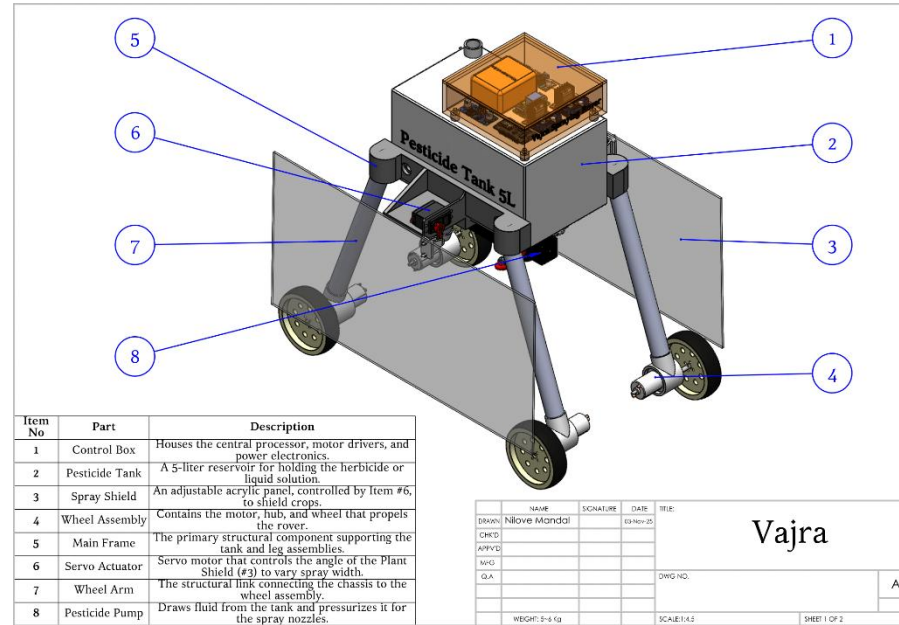
Achievement: Engineered a passive thermal management system that keeps internal temperatures stable even during continuous servo operation.

Project Title: Vajra 2.0: Precision Agrochemical Spraying Platform

A reliability-engineered, teleoperated rover designed to eliminate human exposure to hazardous pesticides.

Problem Statement:

Manual pesticide application accounts for significant farmer health risks (inhalation/dermal contact) and environmental waste due to inefficient broadcast spraying.

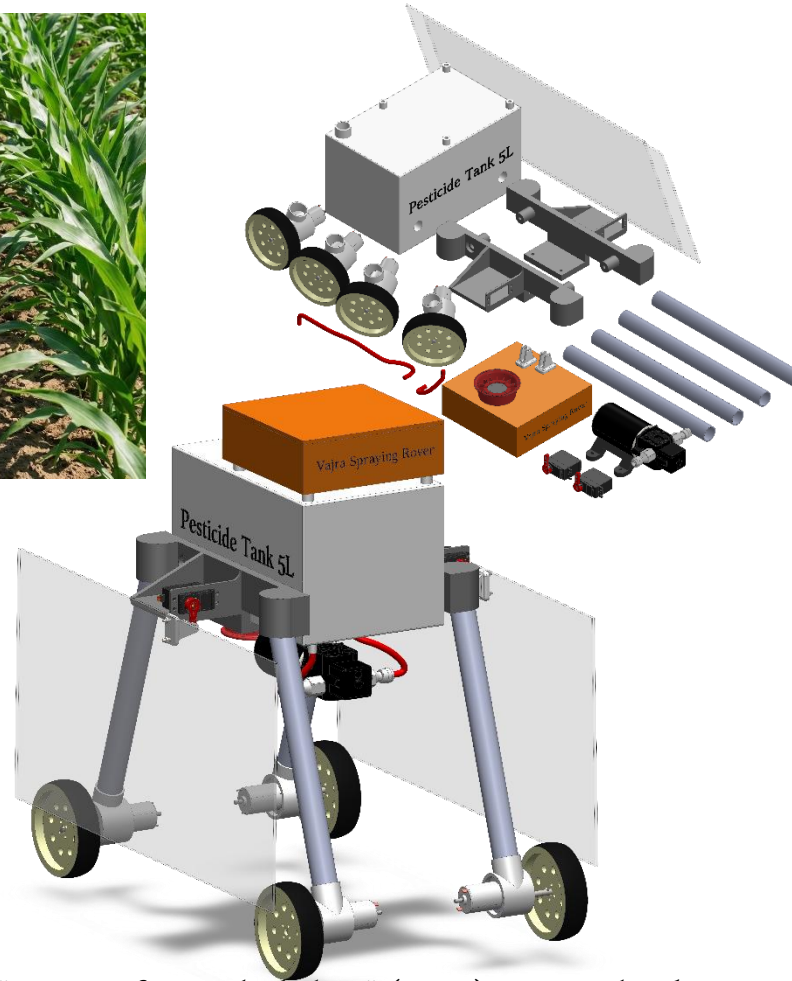
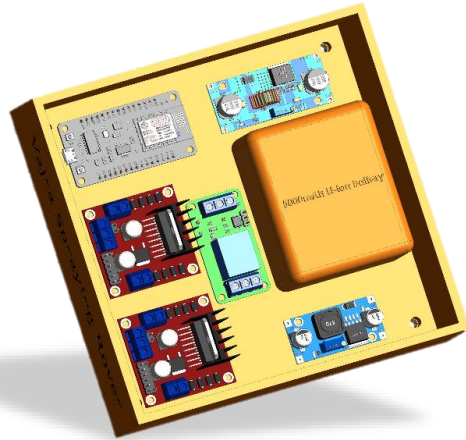


A robust 5-6kg four-wheel-drive rover engineered to carry a 5-liter liquid payload. Unlike typical suspension rovers, Vajra 2.0 utilizes a rigid **Fixed-Arm Chassis** angled at 110° to lower the centre of gravity and counteract the destabilizing "sloshing" forces of the fluid tank.

Key Achievement:
Solved the "Liquid Slosh Resonance" instability common in small AGVs by designing a rigid geometry that locks the fluid mass relative to the drive train, ensuring predictable handling on uneven terrain.

Solution/Impact: Acts as a teleoperated proxy, allowing farmers to spray hazardous chemicals from a safe distance, reducing direct exposure risk by 100%.

Drift Control & Modular Architecture



About Features:

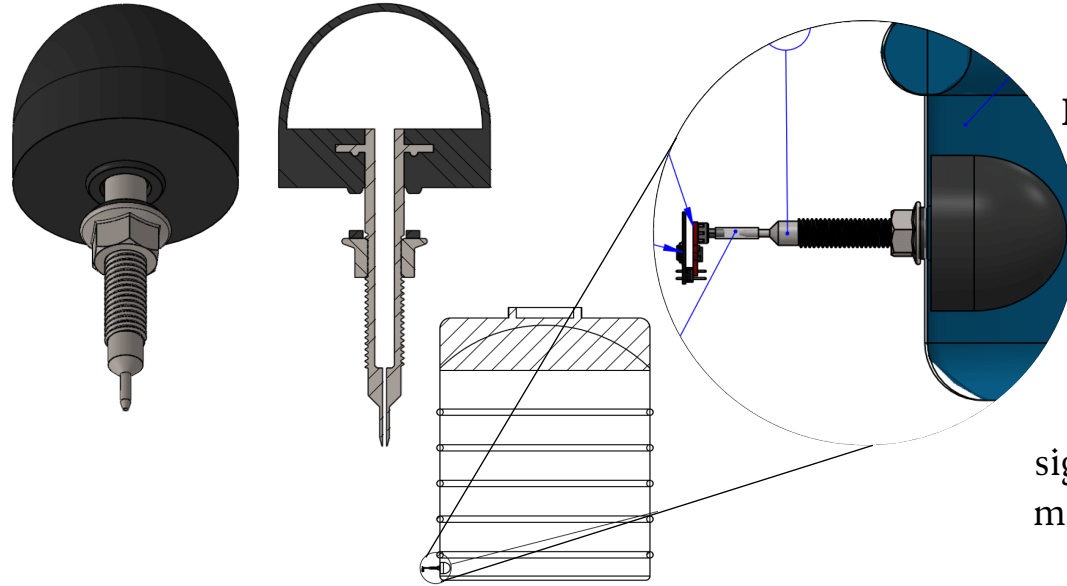
- **Active Spray Shield:**
Incorporates a mechanical shielding system to contain spray mist, preventing wind drift and ensuring chemicals land only on the target crop rows.
- **Modular "Brain" Architecture:** Designed with a split-compute architecture. An ESP32 handles real-time motion control (hard real-time), while the system is pre-wired to accept a Raspberry Pi 4/5 for future computer vision tasks (crop row detection/weed identification) without redesigning the power train.

Key Achievement: Validated "Design for Reliability" (DFR) principles by prioritizing chassis stiffness and modular repairability over complex suspension systems.

Project Title: HydroGauge: Pneumatic Bladder Level Sensor

A non-contact hydrostatic sensor designed to eliminate electronic failure points in liquid storage tanks.

Problem Statement:
Traditional immersion sensors corrode in harsh chemicals, while ultrasonic sensors struggle with foam and condensation in closed tanks.



This system uses a submerged butyl rubber bladder coupled pneumatically to an external pressure sensor. By isolating the electronics outside the tank and using an inline hydrophobic filter/condensate trap, the system prevents liquid ingress and sensor drift caused by humidity.

Key Achievement:
Engineered a temperature-compensation algorithm that linearizes bladder elasticity offset, significantly improving measurement accuracy over standard hydrostatic probes.

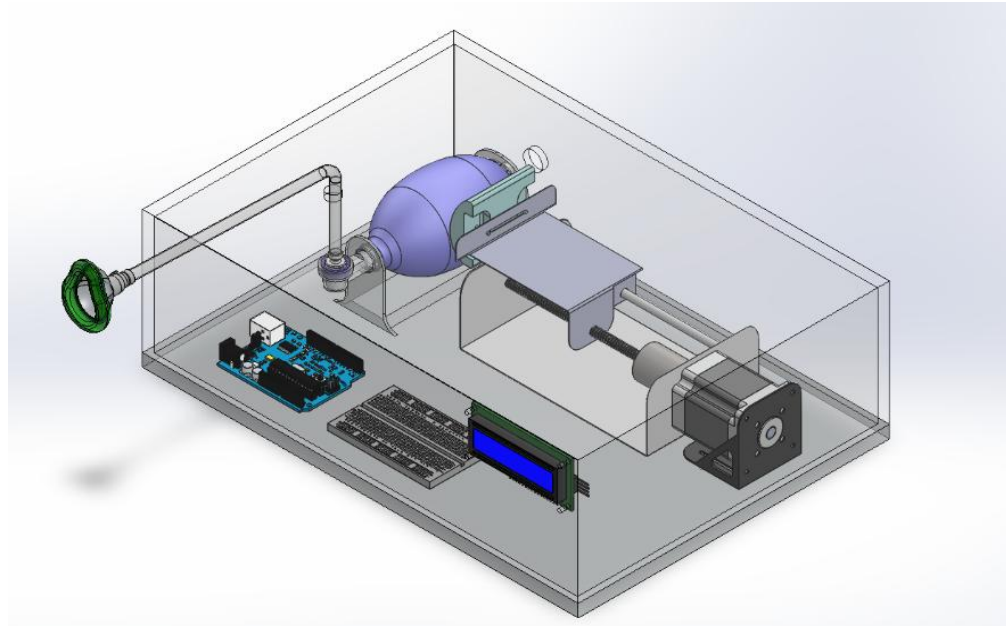
Solution/Impact: Created a retrofittable, low-cost solution that allows for precise level monitoring in opaque or hazardous liquid tanks without exposing sensitive electronics to the fluid.

Smart Breathe: Low-Cost Adaptive Ventilator with Telemetry

Democratizing critical respiratory care with an INR 8,500 IoT-enabled ventilator.

Problem Statement:

High-end ventilators (like the Puritan Bennett™ 840) are expensive, complex to operate, and lack portability, creating a critical gap in resource-limited healthcare settings.



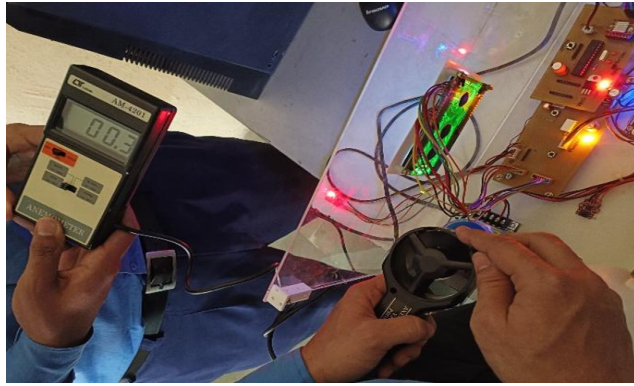
A portable, low-cost mechanical ventilator designed for emergency and home-care use. It automates a silicon Ambu bag (1600ml) using a Nema-17 stepper motor and lead-screw mechanism. Unlike static pump systems, it features an intelligent feedback loop that adjusts airflow based on real-time patient vitals.

Key Achievement: Winner of multiple national innovation awards:

- 1st Prize (Best Innovator): HISET-2024 International Conference at KVVDM Karad.
- 2nd Prize (₹15,000): IIT Ropar's "IMPACT 3.0" International Event.

Solution/Impact: Validated a cost-effective design (approx. ₹8,500) that provides "Tele-ICU" capabilities, allowing doctors to remotely monitor patient SpO2 and pressure data from anywhere

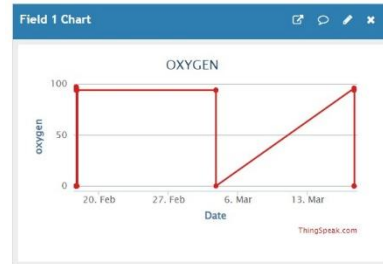
Adaptive Flow Control & IoT Architecture



- I. Flow Rate $\div$ 0.3 to 0.4 m/s (90 – 95) oxygen flow
- II. Flow Rate $\div$ 0.2 m/s (95 – 100) oxygen level

DATA MONITORING USING IOT:
Health parameters that can be monitored are as follows:

- Oxygen level of patient
- Heart rate of patient
- Air Pressure at delivery side of ventilator
- Temperature of patients body.



About Features:

- **Bio-Feedback Loop:** The system reads SpO2 levels via a MAX30100 sensor. If SpO2 drops below 90%, it automatically increases the stroke rate from 10-12 to 15-20 strokes/min to boost oxygenation.
- **Remote Telemetry:** An ESP32 microcontroller transmits patient data (SpO2, Pulse, Air Pressure, Temp) to a cloud dashboard (ThingSpeak) every 60 seconds for trend analysis.

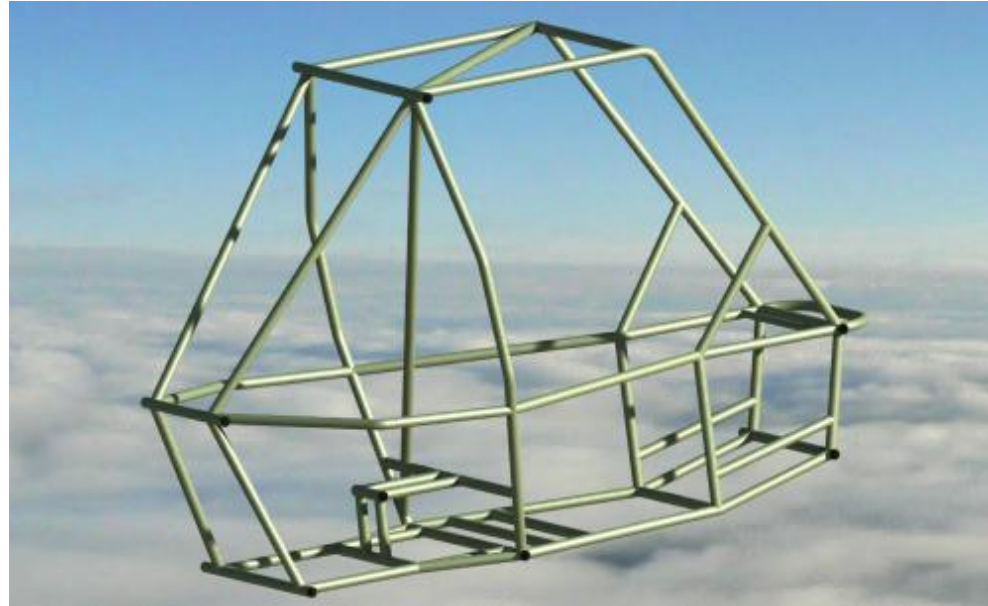
Key Achievement: Validated air discharge of **4.28 Liters/min**, meeting the respiratory requirements for COPD and Asthma support.

Project Title: Team Avibhuj: SAE ATV Roll Cage Design

Optimizing driver safety and chassis stiffness for an all-terrain competition vehicle.

Problem Statement:

Designing a vehicle chassis that is lightweight enough for racing yet strong enough to withstand 10ft drop impacts and frontal collisions.



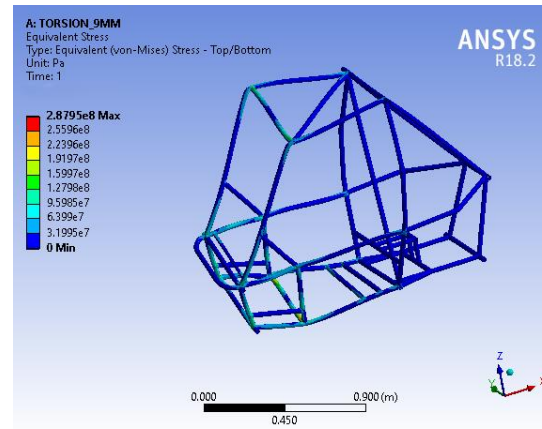
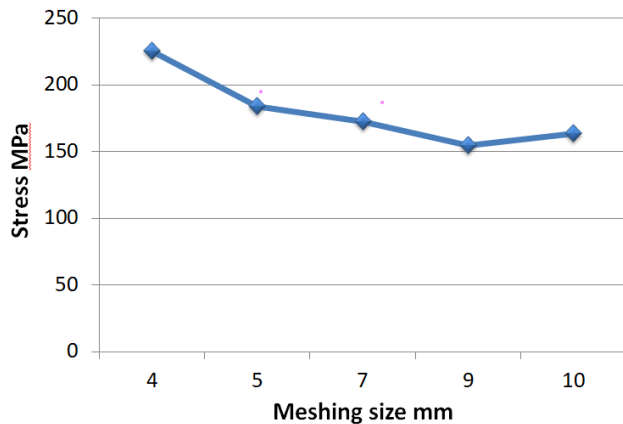
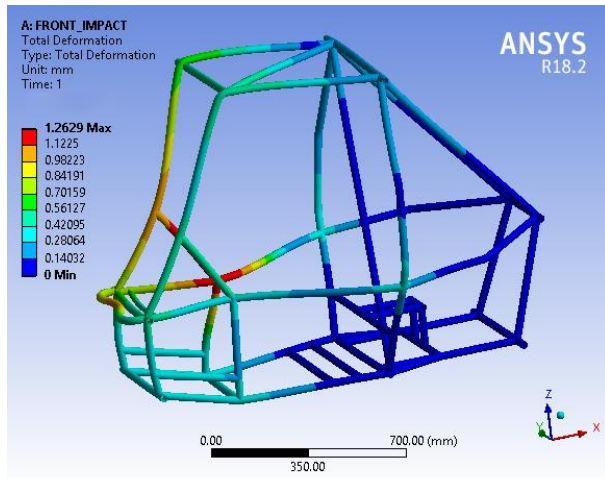
Key Achievement:

Achieved a Safety Factor of 2.97 for Front Impact (13,500N load) while maintaining a chassis weight of only 30kg.

Lead the CAE (Computer-Aided Engineering) evaluation for the roll cage. Utilized CATIA for surfacing and ANSYS Workbench for static and explicit dynamic analysis. We analysed front impact, side impact, and rollover scenarios to ensure compliance with ATVC 2024 Rulebook safety standards.

Solution/Impact: The optimized AISI 4130 tubular space frame provided superior torsional stiffness (preventing suspension flex transfer) and ensured driver survival space in high-speed crash simulations.

Finite Element Analysis (FEA) & Optimization



About the Special Features:

- **Mesh Convergence:** Utilized QUAD/TRIA shell elements for 2D meshing to balance computational efficiency with result accuracy.
- **Modal Analysis:** Calculated natural frequencies (1st mode at 45.4 Hz) to ensure the frame would not resonate with the engine vibration frequency (15-25 Hz), preventing structural fatigue.

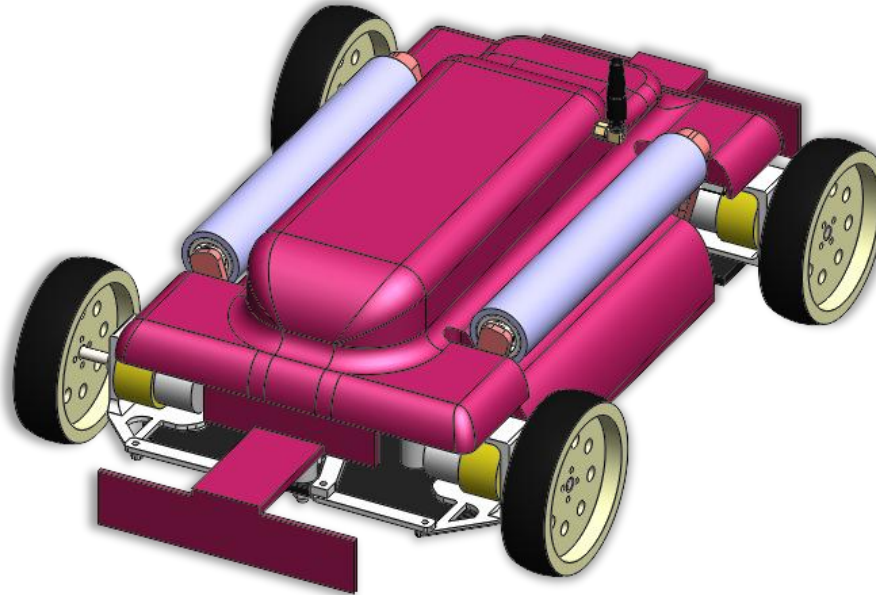
Key Achievement: Verified Torsional Stiffness > 3.5 kNm/degree, ensuring the suspension works effectively without chassis twisting.

Project Title: Trinetra V-Clean: Autonomous Solar Panel Cleaning Robot

A vision-guided robotic solution designed to restore peak efficiency to solar arrays using novel multidirectional manoeuvrability.

Problem Statement:

Accumulation of dust and bird droppings on solar panels can reduce energy output by up to 25%. Existing cleaning robots often struggle to navigate the gaps and irregular layouts of complex solar farms without manual repositioning.



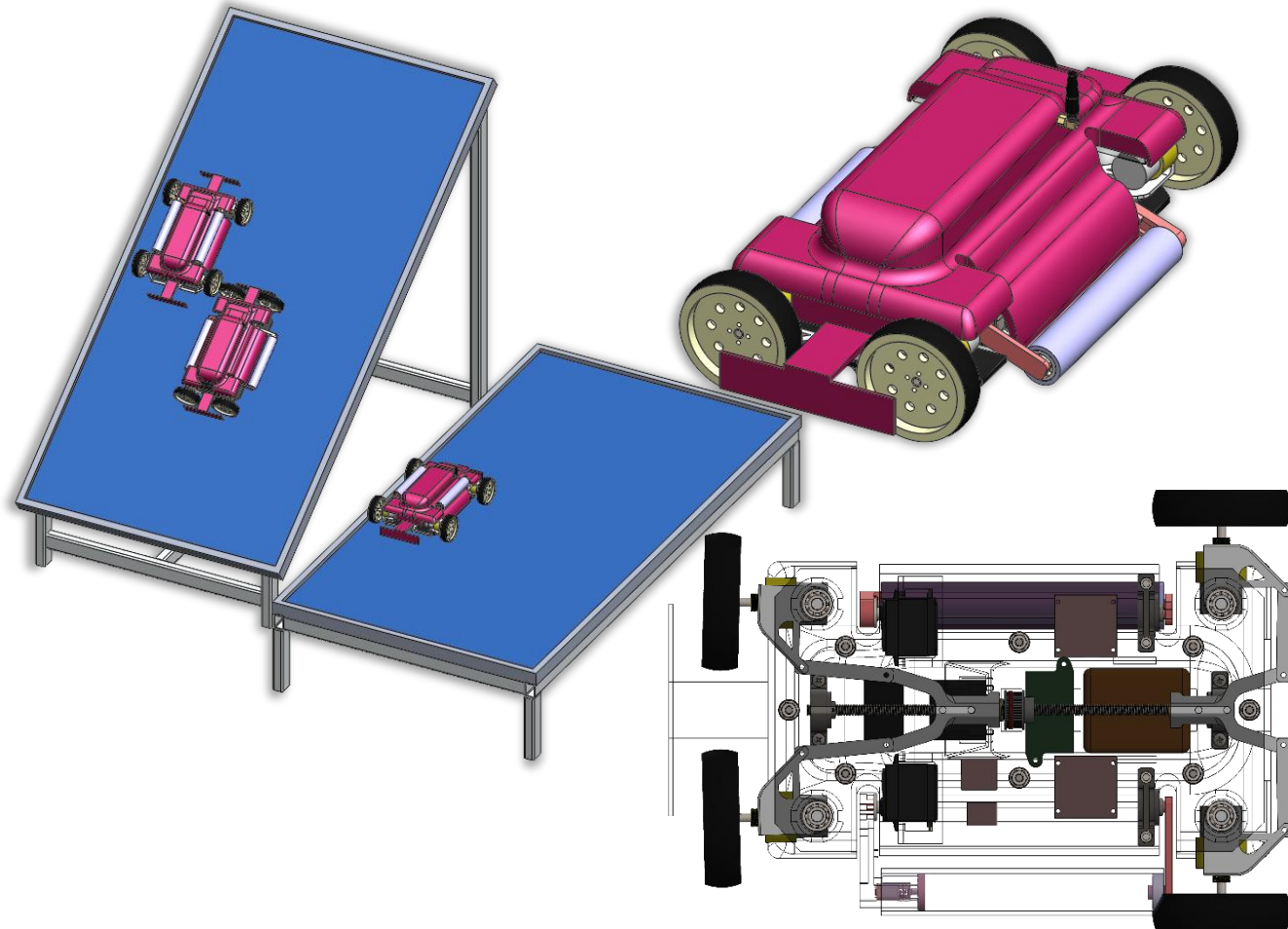
A conceptual design for an autonomous maintenance robot featuring an innovative "Pivoting Wheel" drive system. Unlike standard tracked robots that skid-steer, this design allows for holonomic-style movement, enabling the robot to rotate 90 degrees on the spot to transition between adjacent panels in tight or complex array layouts.

Key Achievement:

Designed a unique pivoting drive train mechanism that decouples wheel orientation from chassis direction, allowing for superior manoeuvrability on narrow panel edges..

Solution/Impact: Automating the cleaning process ensures panels operate at maximum capacity, significantly improving the ROI of solar installations while removing humans from dangerous rooftop environments.

Special Features Text: Vision-Based Navigation & Omni-Directional Mobility



About Features:

- **Pivoting Wheel Drive:**
The core mechanical innovation is a linkage system that pivots the wheel hubs, allowing the robot to "crab walk" or turn with zero radius. This minimizes the risk of the robot falling into the gaps between panels during edge transitions.
- **Vision Guidance System:**
Designed to utilize camera-based edge detection to identify the boundaries of solar panels and detect "hotspots" of dirt, ensuring targeted cleaning rather than random patterns.

Key Achievement: conceptualized a "Complex Array Navigation" logic that allows the robot to clean non-rectangular or segmented solar installations with precision.

Project Title: IoT-Enabled Precision Seeding Unit

Optimizing crop yield potential by strictly enforcing soil-moisture conditions before seeding.

Problem Statement:

Sowing seeds in soil with inadequate moisture leads to poor germination rates and wasted resources.



An automated seeding rig that acts as a "Gatekeeper" for planting. It inserts a moisture probe into the soil; if the moisture level is below the threshold, it halts operation and alerts the farmer via Wi-Fi. If levels are optimal, it proceeds to sow seeds in batches.



Key Achievement:
Programmed a "Batch-Check" algorithm that re-verifies soil conditions after every 10 seeds, ensuring field heterogeneity doesn't compromise the planting batch.

Solution/Impact: Prevents "dead seeds" (failed germination) by ensuring 100% of planted crops are introduced to a viable growth environment.

Project Title: Smart AC Phase Switcher & Load Balancer

An automated arbitration system for fair power distribution in shared-resource environments.

Problem Statement:

In shared housing or multi-meter setups, shared loads (like water pumps) often cause disputes over whose electricity supply is being used, or fail to run if one phase is down.



A solid-state switching unit that monitors AC supply availability from multiple households. It automatically routes the load (pump) to the appropriate power source based on pre-defined priority logic and voltage health.

Key Achievement:

Successfully deployed and verified the system in a live multi-house setup, handling inductive loads (pumps) without contact welding or logic failure.

Solution/Impact: Eliminated manual changeover switches and ensured fair electricity billing by automating the source selection based on usage quotas or availability.

Project Title: Cloud-Synced Biometric Attendance Node

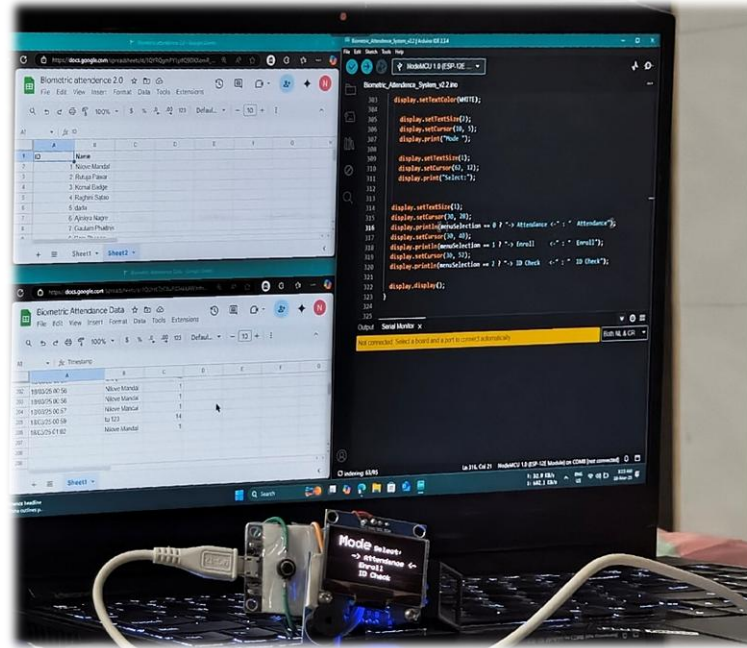
A Wi-Fi-enabled fingerprint logger that eliminates manual data entry by syncing directly to Google Sheets.

Problem Statement:

Traditional attendance systems are either offline (requiring manual data transfer) or expensive proprietary solutions that don't integrate easily with common office tools.

Key Achievement:

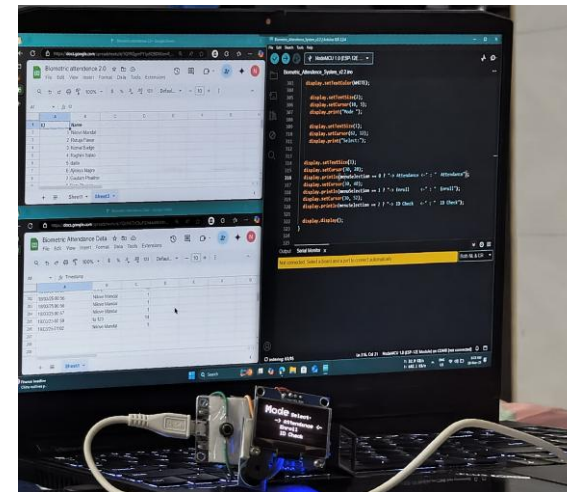
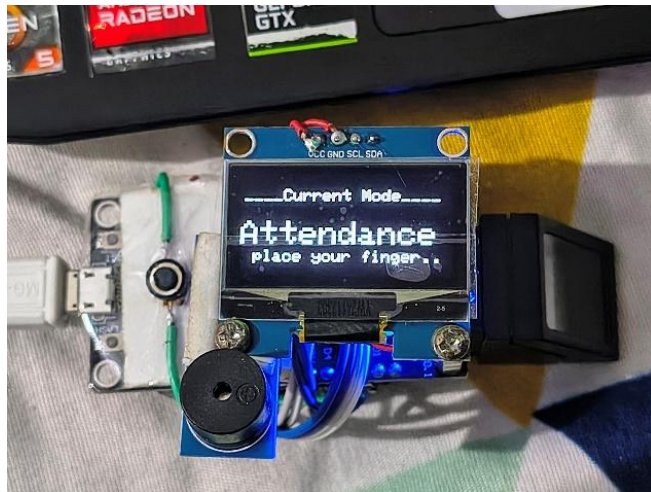
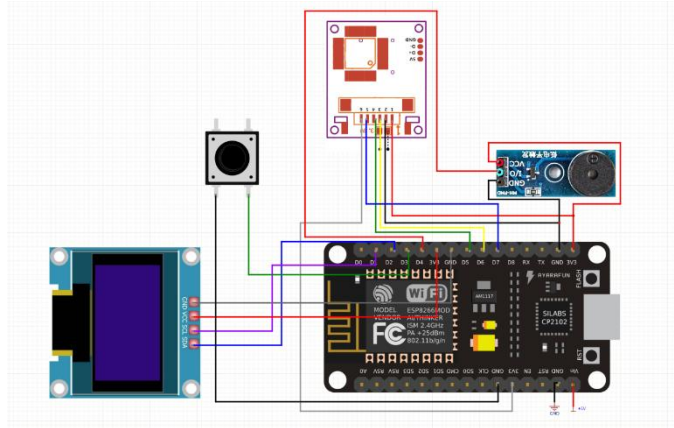
Hacked the Google Sheets API using Google Apps Script to allow an ESP8266 microcontroller to append rows directly to a secure spreadsheet without an intermediate computer.



This device scans a user's fingerprint, verifies their identity locally, and immediately pushes the timestamp and user ID to a cloud-based Google Sheet via Wi-Fi. It bridges the gap between physical hardware security and cloud data management.

Solution/Impact: automated the administrative task of time-logging, reducing payroll processing errors and eliminating "buddy punching" through biometric verification.

IoT Architecture & Data Security



About Features:

- **Direct-to-Cloud**
Logging: Unlike systems that save to an SD card, this system uses an HTTPS request to trigger a script that populates the spreadsheet in real-time.
- **Feedback System:**
Features an OLED display and buzzer to give immediate user feedback (Success/Fail) based on server response codes.

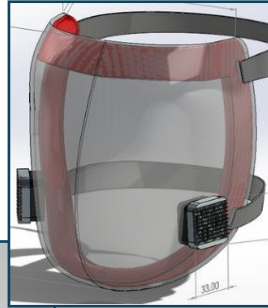
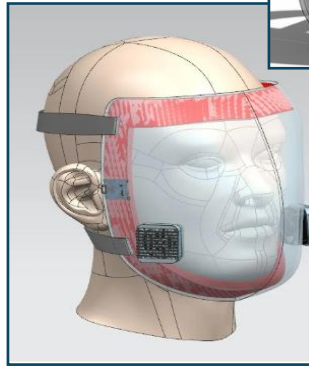
Key Achievement: Reduced attendance logging time to <2 seconds per user with 99% reliable cloud sync.

Project Title: Optically Optimized Anti-Fog Face Shield

Redesigning PPE for extended wear comfort and optical clarity in high-humidity medical environments.

Problem Statement:

Standard medical face shields suffer from rapid fogging and optical distortion, causing eye strain and reducing visibility for healthcare workers during critical procedures.



A research-driven design project focused on improving the ergonomics and optical properties of protective equipment. The design features a calculated curvature to minimize visual parallax and integrates ventilation channels to promote airflow, reducing fog buildup.



Key Achievement:

Conducted comparative research on material coatings and shield geometry to propose a design that maintains 95%+ visible light transmission while mitigating breath condensation.

Solution/Impact: Addressed the "tunnel vision" and fogging issues common in low-cost PPE, providing a design that remains clear during extended periods of physical exertion.

Thank You

I look forward to further discussing with you how I can contribute to the goals of your organization.

Please contact me at your earliest convenience via email at
nilovemandal@gmail.com or by phone +91 93074 74959